Oregon State University — College of Health

H524: Introduction to Biostatistics (AI-Integrated)

Term: Fall 2025
Credits: 4
Delivery: Online (Canvas + Zoom)
Lecture: Mon/Wed 12:00–1:20 pm (PT)
Lab: Wed 8:00–9:50 am (PT)
Instructor: Dr. John Molitor
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Email: john.molitor@oregonstate.edu
Office Hours: Wed 10:00–11:30 am (or by appt.)

Course Description

This course introduces quantitative analysis and interpretation of health data. Topics include data visualization and summary measures; probability and common distributions; sampling distributions; confidence intervals; hypothesis testing; comparisons of means and proportions (including nonparametrics); one-way ANOVA; correlation and simple linear regression; and sample size and power.
AI-integrated focus: You will learn to complete the above with—and because of—generative AI (GenAI), treating AI as a modern calculator and research partner. You'll practice prompt design, critique model outputs, validate results statistically, and document responsible use.

Prerequisites

Basic statistical knowledge at the level of an undergraduate statistics course and MTH 105/111 or higher.

Required & Allowed Tools

Required (provided by OSU): - **RStudio** (desktop or server) with **Microsoft Copilot/Copilot Chat** enabled for R/R Markdown/Quarto. - **R (latest stable)** with tidyverse and base packages used in the course.

Allowed (optional): ChatGPT, Claude (Anthropic), and any other GenAI tools you personally have access to. Use is welcome **and** evaluated—see "AI Use Policy & Documentation."

Text & Notes: Pagano & Gauvreau, *Principles of Biostatistics* (2nd ed.)—recommended; weekly instructor notes and R tutorials are provided in Canvas.

Measurable Student Learning Outcomes

By the end of the course you will be able to: 1. Select and generate appropriate graphical and numerical summaries of data. 2. Use principles of statistical inference to make conclusions about populations from samples. 3. Communicate statistical findings effectively to technical and non-technical audiences. 4. Use computer software (R) to conduct statistical analysis.

> **AI overlay:** For each outcome, you will demonstrate the ability to (a) elicit useful AI assistance, (b) verify and diagnose errors, and (c) document and attribute AI contributions.

Course Flow (Topics by Week)

The content sequence mirrors our traditional H524 flow; each week adds an AI task that deepens both statistics and AI skills.

- **Week 0 (Getting Started):** Intro to R/RStudio, data presentation, numerical summaries.
  AI task: Configure Copilot in RStudio; craft prompts to generate EDA code, then hand-refine; compare Copilot vs ChatGPT on ggplot prompts; write a brief "AI pitfalls I saw" memo.
- **Week 1:** Data summaries continued; rates & standardization.
  AI task: Ask two models to explain age standardization with a worked R example; reconcile differences; produce a verified RMarkdown with citations and a short validation checklist.
- **Week 2:** Probability.
  AI task: Use GenAI to derive/binomial and normal examples; verify with `dbinom`/`pbinom` & `dnorm`/`pnorm`; log any hallucinations and corrections.
- **Week 3:** Probability distributions; sampling distributions (means/proportions).
  AI task: Have Copilot scaffold a simulation to illustrate the CLT; stress-test with skewed distributions; submit a simulation report.

- **Week 4:** Confidence intervals.
  AI task: Prompt three tools to produce a 95% CI workflow for means and proportions; compare intervals & assumptions; submit a "triangulation table."

- **Week 5:** Hypothesis testing (one- and two-sided); **Midterm (see Assessments).**
  AI task: Build a prompt that yields both NHST and CI-based decisions; critique the AI's p-value narrative for practical vs statistical significance.

- **Week 6:** Power; comparing two means; one-way ANOVA.
  AI task: Use AI to write power analysis code; validate with `power.t.test` and a small simulation; document discrepancies.

- **Week 7:** Nonparametric methods; two-sample proportions (risk difference, risk ratio, odds ratio).
  AI task: Prompt AI for RD/RR/OR formulas and R code; confirm with hand calculations on a 2×2 table; reflect on interpretation differences AI produced.

- **Week 8:** Contingency tables; multiple 2×2 tables (confounding/effect modification); catch-up.
  AI task: Use AI to propose stratified analyses; verify with `epitools` or base R; write an AI-informed but student-verified interpretation.

- **Week 9:** Correlation & simple linear regression.
  AI task: Compare Copilot vs ChatGPT to: (1) generate diagnostic plots and (2) state SLR assumptions; stress-test on outliers and nonlinearity.

- **Week 10:** Sample size & power; course synthesis.
  AI task: Co-design a study with an AI assistant; produce a short preregistration outline and power justification; prepare group project presentations.

Finals Week: Final exam window and **Group Project presentations**.

Assessments & Grading

- **AI-Integrated Problem Sets (weekly, individual): 30%**
  Each set includes: (a) core statistical problems; (b) an **AI comparison mini-task** (e.g., Copilot vs ChatGPT vs Claude); and (c) an **AI Use Appendix** with prompt snippets, model/version, and verification notes.

- **Midterm Exam (Week 5, individual, open-resources/AI with verification): 25%**
  Timed take-home. You may use AI, but you must (1) state when you used it,

(2) provide your verification steps, and (3) resolve any contradictions. Pure copy-paste without validation will not earn credit.

- **Final Exam (Finals Week, individual, open-resources/AI with verification): 25%**
  Cumulative, applied analyses in R. Includes short-answer interpretation and code-debugging. AI use is documented and checked against your outputs and reasoning.

- **Participation (discussion, labs, peer review): 10%**
  Includes prompt-critique threads, short lab check-ins, and feedback on peers' AI strategies.

- **Group Project (see below): 10%**
  Team deliverable + recorded/verbal presentation focusing on **how** AI was used to learn, analyze, and communicate.

Grading scale: Standard OSU letter grades. See Canvas for cutoffs and rounding policy.

Group Project (AI Use, Transparency, and Reproducibility)

Teams: 3–4 students.
Data: Choose from instructor-approved public health datasets (or propose your own).
Deliverables:
1) **Reproducible R project** (Quarto/Rmd) covering EDA, an appropriate method from H524 (e.g., CI/HT, two-group comparison, ANOVA, or SLR), assumptions/diagnostics, and clear interpretation.
2) **AI Methods Appendix** with (a) tools used; (b) key prompts (can redact identifiers), (c) what changed after verification; (d) *model comparison note*—what Copilot did better/worse than ChatGPT/Claude; (e) risk-mitigation steps.
3) **10-minute presentation** (live or recorded) that *verbalizes* how you used AI to self-teach, debug, and decide among methods. Include 1–2 "AI fail" examples and how you corrected them.

Rubric (high-level):
- *Statistical correctness & scope* (35%)
- *AI use quality & transparency* (25%)
- *Verification & reproducibility* (20%)
- *Communication (visuals, clarity, audience fit)* (20%)

AI Use Policy & Documentation

- **AI is required** for labs and problem sets; optional for exams but allowed.

- **Always verify.** Numerical results and code must be *replicated* by you in R, with your reasoning shown.

- **Document responsibly.** Keep an **AI Use Appendix** for graded work (tool name & version/date, prompt snippets, your checks).

- **Originality & integrity.** Do not submit AI output as your own analysis without verification, and do not use AI to fabricate data, references, or diagnostics. Using AI without disclosure or with falsified verification is an academic integrity violation.

- **Privacy & data security.** Use only approved/public data with public models. Do not upload restricted or sensitive data to external AI tools.

- **Attribution.** When text is substantially AI-generated (e.g., code comments or draft explanations), add a short attribution in the appendix (e.g., "Drafted with Copilot; revised by Student; verified against R output").

Learning Resources & Support

- Weekly narrated/expanded notes and R tutorials in Canvas.

- Quick-reference prompt patterns for statistics (provided in each unit).

- Links to R resources (UCLA, R4DS, swirl, etc.) and safe-use AI guides.

- Discussion forum monitored by the instructional team.

Exams (Format & Expectations)

- **Open resources/AIs** are permitted; collaboration with people is **not**.

- You must include a short **AI Use Appendix** and **Verification Checklist**.

- We grade reasoning and correctness—not model eloquence. Contradictory or unverified AI claims lose credit.

Lab Structure (Wednesdays)

1. **Setup & Prompting** (Weeks 0–2): Configure tools; structured prompt exercises for EDA and probability; reproducibility with Quarto.
2. **Inference Workflows** (Weeks 3–5): CIs and tests for means/proportions; simulation to validate assumptions; AI critique.
3. **Comparisons & ANOVA** (Week 6–7): Effect sizes, multiple comparisons; nonparametrics; model-suggestions from AI vs your decisions.
4. **Tables, Regression, Power** (Weeks 8–10): Stratified analyses; SLR diagnostics; sample size & power with AI + R verification.

Course Policies (abridged)

- **Academic Integrity:** OSU Student Conduct Code applies. AI use must be disclosed and verified; undisclosed/unchecked AI use is misconduct.

- **Accessibility:** Accommodations via DAS; please contact early.

- **Religious accommodations:** Available per OSU policy.

- **Diversity & Inclusion:** We aim for an affirming, respectful learning environment.

- **Late Work:** Problem set solutions post quickly; late submissions generally not accepted unless pre-arranged.

Weekly Submission Pattern (typical)

- **Mon/Wed:** Mini-checks in lecture; prompt/critique posts as assigned.

- **Wed Lab:** In-lab Copilot task with verification notes.

- **Fri:** Problem set due (with AI Use Appendix).

- **Ongoing:** Project milestones in Weeks 6, 8, 10.

Questions?

Post in the Canvas Q&A forum so everyone benefits. For private matters, use Canvas Inbox or email.

Weekly Notes & AI-Integrated Outline

Week 0 — Orientation, R, & Responsible AI Use

- **Stats focus:**
    - Course overview; types of variables; samples vs. populations; the inference problem.
- **R/Copilot setup:**
    - Install/verify R, RStudio; sign in to Copilot; enable extensions in RStudio.
    - Tour Projects, Quarto, version control basics (optional).
- **AI practice:**
    - Prompt hygiene (roles, constraints, data context); reproducible prompting in Quarto.
    - Pair-programming with Copilot; citing/attributing AI; integrity, privacy, data sensitivity.
- **Mini-lab:**
    - Import a CSV; generate/copilot-scaffold basic summaries; hand-refine code and captions.
- **Deliverables:**
    - AI usage log #1; micro-reflection (3–5 sentences).

Week 1 — Data Presentation & Numerical Summaries (+ Rates/Standardization)

- **Stats focus:**
    - Tables; histograms/density; boxplots; bar charts; mean/median/SD/IQR; rates; age-standardization.
- **R/Copilot skills:**
    - readr, dplyr, ggplot2 pipelines; Copilot plot scaffolds; Quarto chunk options.
- **AI practice:**
    - Ask Copilot/ChatGPT/Claude for visualization choices; justify final selection.
- **Homework:**
    - Data brief with ≥2 plots; include AI excerpts and rationale for edits.

Week 2 — Probability Foundations

- **Stats focus:**
    - Set rules; conditional probability; Bayes; probability by simulation.
- **R/Copilot skills:**

- sample(), runif(), rbinom() workflows; Copilot-generated annotations.
- **AI practice:**
  - Solicit analogies from each model; rank clarity/correctness; correct hallucinations.
- **Homework:**
  - “Monte Carlo postcard”—simulate a probability scenario; compare Copilot vs. ChatGPT code.

Week 3 — Theoretical Distributions & Sampling Distribution

- **Stats focus:**
  - Binomial; Normal (68-95-99.7); t; CLT; standard error; quantiles.
- **R/Copilot skills:**
  - dbinom/pbinom/qbinom, dnorm/qnorm, rt/qt; inline explanations.
- **AI practice:**
  - Have AI choose distributions for 3 scenarios and defend choices.
- **Homework:**
  - CLT simulation mini-lab + quiz; include prompts that improved code.

Week 4 — Confidence Intervals (One Population)

- **Stats focus:**
  - CIs for mean and proportion; interpretation; margin of error; optional bootstrap.
- **R/Copilot skills:**
  - Build a CI helper function; produce a CI summary table.
- **AI practice:**
  - Prompt 3 models for lay explanations; select/revise best.
- **Homework:**
  - CI memo on a public-health dataset; include provenance and checks.

Week 5 — Hypothesis Testing (One Population) + Midterm

- **Stats focus:**
  - H0/HA; z/t test statistics; p-values; Type I/II errors; practical vs. statistical significance.
- **R/Copilot skills:**
  - t.test() scaffolds; Quarto with inline results; decision rules.
- **AI practice:**
  - Each model proposes a full workflow; you critique and implement.
- **Assessment:**
  - Midterm (individual, open resources/AI with verification appendix).

Week 6 — Power, Two Means, One-Way ANOVA

- **Stats focus:**
  - Power concepts; independent/paired t-tests; ANOVA; post-hoc comparisons.
- **R/Copilot skills:**
  - power.t.test(), aov(); diagnostic plots; assumption checks.
- **AI practice:**
  - Model-recommended tests for 4 vignettes; verify with R.
- **Homework:**
  - Design-a-study worksheet (choose n/detectable effect) with AI reasoning vs. final plan.

Week 7 — Nonparametric Methods

- **Stats focus:**
  - Wilcoxon signed-rank; Mann-Whitney; robustness; reporting medians/effect sizes.
- **R/Copilot skills:**
  - wilcox.test(); rank-based summaries.
- **AI practice:**
  - Diagnose when parametric assumptions fail; compare to visual diagnostics.
- **Homework:**
  - Parallel analyses (t-test vs. Wilcoxon) with trade-off discussion.

Week 8 — Contingency Tables & Multiple 2×2s

- **Stats focus:**
  - Two-proportion comparisons; RD, RR, OR; chi-square; Mantel–Haenszel; confounding & effect modification.
- **R/Copilot skills:**
  - prop.test(), fisher.test(); epiR/oddsratio utilities; table formatting.
- **AI practice:**
  - Compute OR/RR with CIs; explain rare-disease approximation for lay audience.
- **Homework:**
  - Case-control vs. cohort mini-cases; reconcile model outputs; cite decision rules.

Week 9 — Correlation & Simple Linear Regression

- **Stats focus:**

- Scatterplots; covariance & Pearson r; SLR estimation; slope inference; CIs/prediction; diagnostics (intro).

- **R/Copilot skills:**
  - lm(), summary(), predict(); model report sections.
- **AI practice:**
  - Submit a misleading AI regression claim you encountered and how you corrected it.
- **Homework:**
  - SLR analysis on the provided dataset; include prompts and evaluation.

Week 10 — SLR Wrap-Up & Capstone Launch

- **Stats focus:**
  - Interpretation; prediction vs. estimation; residual checks; optional transformations.
- **R/Copilot skills:**
  - Publication-quality regression report via Copilot scaffolds.
- **AI practice:**
  - Capstone blueprint review; responsible/reproducible AI plan.
- **Deliverables:**
  - Team capstone proposal with AI work plan + risk-mitigation strategies.

End-of-Term Group Project (AI Accountability & Verbalization)

- **Goal:** Apply course methods to a public-health question using R with GenAI support; demonstrate reasoning and responsible AI use.
- **Deliverables:**
  - Quarto technical report (EDA, chosen H524 methods, assumptions/diagnostics, interpretation).
  - **AI provenance appendix** (tools, prompts, versions, what was accepted/edited, why).
  - Slide deck for a 6–8 minute team talk.
  - Live **verbalization** by each member on an AI-accelerated or AI-misleading moment and the quality-control applied.
  - Replication pack (data dictionary, session info, seeds).
- **Rubric (high-level):** correctness; clarity; reproducibility; ethical AI use; reflection depth.

Weekly AI-Integrated Homework Pattern (from Week 1)

- **Each assignment includes:**
  - Core stats task in R (graded for correctness/interpretation).
  - AI comparison mini-task (e.g., Copilot vs. ChatGPT vs. Claude) with a brief triangulation table.
  - Short reflection (what AI got right/wrong; how you validated results).
- **Process requirements:**
  - Maintain an **AI Usage Log** (prompt, tool, version/date, changes you made).
  - Attach relevant excerpts/screens to submissions; ensure all numerical results are replicated in R.